An isometric structural no-other-examples theorem for a minimal Lipschitz composition problem

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Status: partial solution, likely valid. The unrestricted question remains open. Under either of two natural hypotheses on the maximal hull, the result rules out both a new operator class and a nonstandard ideal norm.

1 Source question

Nahuel Albarracin and Pablo Turco ask, on page 11 of *On the Lipschitz operator ideal* $\mathrm{Lip}_0 \circ \mathcal{A} \circ \mathrm{Lip}_0$ [1], whether there is any minimal Banach operator ideal $\mathcal{A}$ other than the approximable ideal $\overline{\mathcal{F}}$ such that

$$(\mathrm{Lip}_0 \circ \mathcal{A}^{\min} \circ \mathrm{Lip}_0)^{\min}$$

is of composition type. Since the question concerns minimal $\mathcal{A}$, one may write $\mathcal{A}^{\min} = \mathcal{A}$. Their Proposition 2.17 proves the necessary class condition

$$(\mathcal{A}^{\max})^{\mathrm{sur}} = \mathcal{L}. \tag{1}$$

In particular, for the ideal of approximable operators, by Proposition 2.13, $\mathrm{Lip}_0 \circ \overline{\mathcal{F}} \circ \mathrm{Lip}_0$ is not of composition type, but by Proposition 2.16 its minimal kernel is of composition type, giving a negative answer of [30, Problem 4.15].

We do not know any other example besides $\overline{\mathcal{F}}$ of a minimal Banach operator ideal such that $(\mathrm{Lip}_0 \circ \mathcal{A}^{\min} \circ \mathrm{Lip}_0)^{\min}$ is of composition type. However, we have the following result. Before stating it, recall that, following [26, Definition 4.7.1], an operator $T\colon E \to F$ belongs to the *surjective hull* of a Banach operator ideal $\mathcal{A}$, denoted as $\mathcal{A}^{\mathrm{sur}}$, if $T \circ Q_E \in \mathcal{A}(\ell_1(B_E), F)$ where $Q_E\colon \ell_1(B_E) \to E$ is the canonical quotient map.

Proposition 2.17. *Let $\mathcal{A}$ be a Banach operator ideal. If $(\mathrm{Lip}_0 \circ \mathcal{A}^{\min} \circ \mathrm{Lip}_0)^{\min}$ if of composition type, then $(\mathcal{A}^{\max})^{\mathrm{sur}} = \mathcal{L}$.*

Figure 1: The exact source passage and its necessary condition, page 11 of arXiv:2211.14240.

2 Definitions

For a Banach operator ideal $\mathcal{B}$, its surjective hull is characterized by

$$T \in \mathcal{B}^{\mathrm{sur}}(E, F) \quad \Longleftrightarrow \quad TQ_E \in \mathcal{B}(\ell_1(B_E), F),$$

where $Q_E : \ell_1(B_E) \to E$ is the canonical quotient. The ideal is *surjective* if $\mathcal{B} = \mathcal{B}^{\mathrm{sur}}$ isometrically. It is *symmetric* if $T \in \mathcal{B}$ exactly when $T^* \in \mathcal{B}$, with equal ideal norms, and *injective* if an isometric embedding $i : F \to G$ satisfies $iT \in \mathcal{B}$ only if $T \in \mathcal{B}$. These are the standard Banach-ideal conventions [3, 4].

3 Idea of the proof

Composition type carries more norm information than (1): for every finite-dimensional E it forces

$$\|I_{\mathcal{F}(E)}\|_{\mathcal{B}} = \|I_E\|_{\mathcal{A}}. \tag{2}$$

For $E = \mathbb{R}$, the left side is the $\mathcal{B}$-norm of the identity of $\mathcal{F}(\mathbb{R}) = L_1(\mathbb{R})$, while the right side is one. Every separable Banach space is a metric quotient of a norm-one complemented copy of ℓ_1 in this L_1 space. This first upgrades (1) to the isometric equality $\mathcal{B}^{\mathrm{sur}} = \mathcal{L}$; maximality then removes the separability restriction. The surjective case collapses immediately. In the symmetric and injective case, adjoints of the canonical quotient maps give the canonical isometric evaluation embeddings into ℓ_∞ spaces. Injectivity removes these embeddings and forces every identity into $\mathcal{B}$ with norm one.

4 Partial classification theorem

Theorem 1. *Let $\mathcal{A}$ be a minimal Banach operator ideal and put $\mathcal{B} = \mathcal{A}^{\max}$. Suppose that either*

1. *$\mathcal{B}$ is surjective, or*
2. *$\mathcal{B}$ is symmetric and injective.*

Then, isometrically,

$$(\mathrm{Lip}_0 \circ \mathcal{A} \circ \mathrm{Lip}_0)^{\min} \text{ is of composition type} \quad \Longleftrightarrow \quad \mathcal{A} = \overline{\mathcal{F}}.$$

In particular, the source question has neither an additional operator class nor a nonstandard ideal norm in either structural case.

Proof. Assume first that $(\mathrm{Lip}_0 \circ \mathcal{A} \circ \mathrm{Lip}_0)^{\min}$ is of composition type. We begin by retaining the norms in Propositions 2.14–2.15 of [1]. Write $\mathcal{S} = \mathrm{Lip}_0 \circ \mathcal{A} \circ \mathrm{Lip}_0$. For finite-dimensional E, the Dirac map $\delta_E : E \to \mathcal{F}(E)$ factors strongly through I_E, so

$$\|\delta_E\|_{\mathcal{S}} \le \|I_E\|_{\mathcal{A}}.$$

The composition-type hypothesis and Proposition 2.14 give isometrically $\mathcal{S}^{\max} = \mathcal{B} \circ \mathrm{Lip}_0$. Since the composition norm of a Lipschitz map is the $\mathcal{B}$-norm of its linearization,

$$\|I_{\mathcal{F}(E)}\|_{\mathcal{B}} = \|\delta_E\|_{\mathcal{B}\circ\mathrm{Lip}_0} = \|\delta_E\|_{\mathcal{S}^{\max}} \le \|I_E\|_{\mathcal{A}}. \tag{3}$$

On finite-dimensional components, $\mathcal{A}$ and $\mathcal{A}^{\max} = \mathcal{B}$ agree isometrically. The Godefroy–Kalton lifting theorem [5] supplies a linear isometry $U_E : E \to \mathcal{F}(E)$ with $\beta_E U_E = I_E$, where $\beta_E : \mathcal{F}(E) \to E$ has norm one. Hence

$$\|I_E\|_{\mathcal{A}} = \|I_E\|_{\mathcal{B}} \le \|I_{\mathcal{F}(E)}\|_{\mathcal{B}}.$$

Together with (3), this proves (2).

We next strengthen Proposition 2.17 of [1] to

$$\mathcal{B}^{\mathrm{sur}} = \mathcal{L} \quad \text{isometrically.} \tag{4}$$

Indeed, put $E = \mathbb{R}$ in (2). The normalization of a Banach operator ideal and the canonical isometry $\mathcal{F}(\mathbb{R}) \cong L_1(\mathbb{R})$ give

$$\|I_{L_1(\mathbb{R})}\|_{\mathcal{B}} = 1. \tag{5}$$

Let G be separable. Choose a sequence dense in the unit sphere of G, with every tail dense. The associated norm-one map $q : \ell_1 \to G$ is a metric quotient: the usual successive-approximation expansion of a vector gives, for every $\varepsilon > 0$, a preimage a with $\|a\|_1 \le (1+\varepsilon)\|q(a)\|$. Also, disjoint unit intervals give an isometric copy $U\ell_1 \subset L_1(\mathbb{R})$ and the map taking interval averages is a norm-one projection P onto it. Thus $qP : L_1(\mathbb{R}) \to G$ is a norm-one metric quotient. By (5), $qP \in \mathcal{B}$ with $\mathcal{B}$-norm one. The quotient property of the surjective hull now gives

$$\|I_G\|_{\mathcal{B}^{\mathrm{sur}}} = 1. \tag{6}$$

Finally, maximal and surjective hulls commute isometrically [4, 3]. Because $\mathcal{B}$ is maximal, $\mathcal{B}^{\mathrm{sur}} = (\mathcal{B}^{\mathrm{sur}})^{\max}$. For arbitrary $T : E \to F$, every finite-dimensional/cofinite compression of T has separable domain; (6) and the ideal property make its $\mathcal{B}^{\mathrm{sur}}$-norm equal to its operator norm. The maximal-hull formula therefore gives $\|T\|_{\mathcal{B}^{\mathrm{sur}}} = \|T\|$, proving (4).

If $\mathcal{B}$ is surjective, then $\mathcal{B} = \mathcal{B}^{\mathrm{sur}} = \mathcal{L}$ isometrically.

Now suppose instead that $\mathcal{B}$ is symmetric and injective. Fix an arbitrary Banach space E. By (4), $\|I_{E^*}\|_{\mathcal{B}^{\mathrm{sur}}} = 1$. The definition of the surjective-hull norm therefore gives

$$Q_{E^*} : \ell_1(B_{E^*}) \longrightarrow E^* \quad \text{in } \mathcal{B}, \qquad \|Q_{E^*}\|_{\mathcal{B}} = 1.$$

By symmetry,

$$Q_{E^*}^* : E^{**} \longrightarrow \ell_\infty(B_{E^*}) \quad \text{belongs to } \mathcal{B}, \qquad \|Q_{E^*}^*\|_{\mathcal{B}} = 1.$$

Precompose with the canonical embedding $J_E : E \to E^{**}$. The resulting operator is

$$j_E := Q_{E^*}^* J_E, \qquad j_E(x) = (x^*(x))_{x^* \in B_{E^*}},$$

the canonical isometric evaluation embedding of E into $\ell_\infty(B_{E^*})$. Hence $j_E \in \mathcal{B}$ with ideal norm one. Isometric injectivity of $\mathcal{B}$, applied to $j_E \circ I_E$, yields $\|I_E\|_{\mathcal{B}} = 1$.

As E was arbitrary, the ideal property applied to TI_E gives $\|T\|_{\mathcal{B}} \le \|T\|$ for every bounded T. The reverse inequality is part of the Banach-ideal axioms. Thus $\mathcal{B} = \mathcal{L}$ isometrically in the second case.

Since $\mathcal{A}$ is minimal, all equalities in

$$\mathcal{A} = (\mathcal{A}^{\max})^{\min} = \mathcal{L}^{\min} = \overline{\mathcal{F}}$$

are isometric. Conversely, for the standard approximable ideal $\mathcal{A} = \overline{\mathcal{F}}$, Proposition 2.16 of [1] gives

$$(\mathrm{Lip}_0 \circ \overline{\mathcal{F}} \circ \mathrm{Lip}_0)^{\min} = \overline{\mathcal{F}} \circ \mathrm{Lip}_0,$$

which is of composition type. This proves both directions. □

5 Verification and limitations

The new quantitative step is (2). It uses the isometric equalities in source Proposition 2.14 and the Godefroy–Kalton linear lifting. The proof of (4) then supplies the norm information not written explicitly in source Proposition 2.17. A reviewer should check the standard isometric commutation of maximal and surjective hulls and the quotient-hull norm convention.

The theorem does not settle maximal hulls that are nonsurjective and fail symmetry or injectivity. The source's L_1-factorable test case lies outside the surjective branch and is ruled out there by a separate geometric argument. Consequently this packet is a solved structural subcase, not a full answer to the unrestricted existence question.

6 Bounded novelty check

The local result and attempt indexes were searched for arXiv:2211.14240 and the core composition-type terminology. A web search on 22 July 2026 used the exact source sentence and combinations of "minimal Banach operator ideal", "surjective hull", "symmetric", and "injective". It found the source [1] and the preceding Galois-connection paper [2], but no later resolution or statement of this structural theorem. Novelty confidence is moderate-low because the proof is a short consequence of standard operator ideal facts and may be folklore.

7 Human-review recommendation

Recommended verdict: *likely valid partial result.* A reviewer should first check the exact norm identity (2), then the metric-quotient and maximal-hull passage proving (4). No computational verification is needed.

References

[1] N. Albarracin and P. Turco, *On the Lipschitz operator ideal* $\mathrm{Lip}_0 \circ \mathcal{A} \circ \mathrm{Lip}_0$, arXiv:2211.14240, especially Propositions 2.16–2.17 and page 11 (2022; revised 2023).

[2] P. Turco and R. Villafane, *Galois connection between Lipschitz and linear operator ideals and minimal Lipschitz operator ideals*, J. Funct. Anal. 277 (2019), 434–451; arXiv:1807.10786.

[3] A. Pietsch, *Operator Ideals*, North-Holland, 1980.

[4] A. Defant and K. Floret, *Tensor Norms and Operator Ideals*, North-Holland, 1993.

[5] G. Godefroy and N. Kalton, *Lipschitz-free Banach spaces*, Studia Math. 159 (2003), 121–141.